

Mortgage Pricing Model (MKM-MP-001)

Wolfram Oracle Benchmark of the Fair-Value Calculation, after the Fix

MKM Research Labs

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1 Executive Summary

This document benchmarks the Mortgage Pricing Model (MKM-MP-001), which values a flood-exposed residential mortgage from its amortisation schedule, an affordability-based credit spread and a loss given default. Version 0.1 tested the pricer against model document v1.1 and found that the fair value counted the expected loss twice and took the recovery a month late. Both are fixed in the pricer, and the document is corrected as v1.3, which this version tests.

The code computes its documented formulae. Over 360 loans — loan-to-value 0.5 to 1.0, recovery haircuts 0 to 40%, two incomes, two coupons, three terms, discounted at the coupon and at a risk-free rate — the payment, spread, expected cashflows, expected losses and fair value agree with a 50-digit implementation of v1.3's Eqs. 1–24 to 1.7×10^{-14}. Wolfram and the 50-digit reference agree to 2.8×10^{-30} on both values.

The fair value counts each loss once. The fair value is now the present value of the expected cashflows, whose recovery is already net of the loss, and it equals Wolfram's independent pathwise valuation: the expectation, over the month of default, of the payments made before it and the recovery made in it. The loss is taken on the balance owed when the month of default began, so the lender never recovers more than the property fetches. The correction raises fair value on the 53% of loans in the grid that carry a loss given default, by up to 34.6% of par: a GBP 300,000 loan that v1.1 valued at GBP 101,992 is worth GBP 205,812 (Table 2). Expected loss itself is reported as before, now on the opening balance.

The document's claims hold (Table 3). v1.3 corrects those v0.1 found wrong: a 100 bp rise raises a 25-year payment by 8.3–11.9%, within the 8–12% now stated; missing income prices at the top of the spread schedule (37.625% at the baseline), as now stated, the 15% fallback being unreachable; and discounting is described as production does it, at the risk-free rate the loan calculator supplies. Table 7 is reproduced exactly, the expected-loss bound holds, and Wolfram proves the par identity symbolically.

Evidence. 6 checks, outcome: all pass (Table 1).

- Wolfram: 15.0.0 for Mac OS X ARM (64-bit) (May 26, 2026) via live kernel session (wolframclient).
- Code under test, at commit `2234867eb786`.
- Numerics: numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1.
- Generated 2026-10-27 by `python -m tools.wolfram_oracle mp001`.

2 Model Purpose and Scope

2.1 What is benchmarked

`LoanPricer.price_loan` in `src/models/loan/pricer/` (`_pricing.py`, with the spread in `credit.py`), imported as production imports it and called as the loan calculator (`src/routes/_loan_pricing`) calls it.

2.2 What is not

The calibration of the spread schedule (§4.3), the coupon the calculator builds upstream of the pricer, and the portfolio aggregation (§3.9), which is arithmetic. This is a benchmark of the implementation (validation question Q8), not independent validation (Q9).

3 Mathematical Framework

Following the document (v1.3, §3): monthly payment M, balances $B_0, \ldots, B_n$, monthly default probability $h = 1 - e^{-s/12}$, survival $S_i = (1-h)^i$, $\mathrm{PD}_i = S_{i-1} - S_i$, recoverable value $R = (1-\delta)V$, discount factors D_i, and the loss on a default in month i taken on the balance owed when the month began, $\mathrm{LGD}_i = \max(0, B_{i-1} - R)$. Then

$$\mathrm{FV} = \mathrm{PV}_{\mathrm{CF}} = \sum_i D_i\big[M S_i + (B_{i-1} - \mathrm{LGD}_i)\,\mathrm{PD}_i\big], \qquad \mathrm{PV}_{\mathrm{EL}} = \sum_i D_i\,\mathrm{LGD}_i\,\mathrm{PD}_i, \tag{1}$$

with $B_{i-1} - \mathrm{LGD}_i = \min(B_{i-1}, R)$, the recovery.

The pathwise value. Let τ be the month of default, so that $P(\tau = i) = \mathrm{PD}_i$ and $P(\tau > n) = S_n$. On the path where it defaults in month τ, the lender receives M in each month before τ and $\min(B_{\tau-1}, R)$ in month τ; on the path where it never defaults, all n payments. Taking the expectation,

$$\mathbb{E}[\mathrm{PV}] = \sum_{i=1}^{n} D_i\big[M\,P(\tau > i) + \min(B_{i-1}, R)\,P(\tau = i)\big] = \mathrm{PV}_{\mathrm{CF}}. \tag{2}$$

What v1.1 did. v1.1 took the loss on the closing balance, $\max(0, B_i - R)$, so its recovery $R + (B_{i-1} - B_i)$ exceeded R by a month's principal when the loan was under water; and it set $\mathrm{FV} = \mathrm{PV}_{\mathrm{CF}} - \mathrm{PV}_{\mathrm{EL}}$, deducting a loss already inside the recovery.

4 Benchmark Method

The equations above are evaluated with `mpmath` at 50 significant digits, from v1.3 of the document rather than the code; the loan-to-value ratio is taken as the double the code compares against its bands, since a band edge such as 0.95 falls between the exact ratio and its double. Wolfram evaluates, from each input's exact value at 60-digit precision, both the document's fair value and the pathwise value, the latter as an expectation over a `GeometricDistribution` default month with the payments' present values accumulated, and it proves symbolically that the contractual payments discounted at the coupon are worth par.

5 Validation and Backtesting

Table 1: Benchmark checks

Check	Result	Tolerance	Outcome
Monthly payment (Eq. 1): model vs document	2.0×10^{-14}	1.0×10^{-12}	pass
Credit spread (Eqs. 6–13): model vs document	6.4×10^{-14}	1.0×10^{-12}	pass
PV of expected cashflows (Eqs. 19, 22): model vs document	1.7×10^{-14}	1.0×10^{-12}	pass
PV of expected losses (Eqs. 20, 23): model vs document	3.8×10^{-14}	1.0×10^{-12}	pass
Fair value (Eqs. 22, 24): model vs document	1.7×10^{-14}	1.0×10^{-12}	pass
Wolfram agrees with the reference (fair value and pathwise value)	2.8×10^{-30}	1.0×10^{-12}	pass

5.1 The correction

Table 2: Fair value before (v1.1) and after (v1.3) the fix: GBP 300k property, 6% coupon, 25 years, discounted at the coupon

LTV	δ	Income	s (%)	v1.1	v1.3	PV of loss	of par
0.80	0.25	30,000	23.06	230,639	234,697	4,423	+1.7%
0.80	0.25	60,000	3.78	237,986	238,786	880	+0.3%
0.80	0.40	30,000	23.06	170,304	204,398	34,721	+14.2%
0.80	0.40	60,000	3.78	221,043	230,242	9,424	+3.8%
0.90	0.25	30,000	34.94	213,773	240,979	27,927	+10.1%
0.90	0.25	60,000	5.35	254,743	261,999	7,516	+2.7%
0.90	0.40	30,000	34.94	138,483	203,311	65,595	+24.0%
0.90	0.40	60,000	5.35	225,724	247,427	22,088	+8.0%
0.95	0.25	30,000	48.35	195,529	239,250	44,530	+15.3%
0.95	0.25	60,000	7.03	255,737	269,868	14,520	+5.0%
0.95	0.40	30,000	48.35	115,840	199,401	84,379	+29.3%
0.95	0.40	60,000	7.03	218,248	251,064	33,323	+11.5%
1.00	0.25	30,000	59.06	179,117	238,469	60,211	+19.8%
1.00	0.25	60,000	9.17	250,966	274,841	24,406	+8.0%
1.00	0.40	30,000	59.06	97,635	197,727	100,953	+33.4%
1.00	0.40	60,000	9.17	204,922	251,767	47,480	+15.6%

5.2 The model document's claims

Table 3: The model document's claims (v1.3), tested

Claim	Stated	Benchmark finds	Outcome
Eqs. 19, 22, 24	fair value is the PV of expected cashflows, net of loss once	equals the pathwise value to 1.7×10^{-14}	met
Eqs. 18–19	the recovery is $\min(B_{i-1}, R)$	never exceeds the recoverable value	met
§6, Table 7	spread and value by flood category	reproduced to the figures quoted; no loss given default at the baseline	met
§6.1	+100 bp raises the 25-year payment 8–12%	8.3–11.9% over rates of 2–9%	met
§3.2.4	missing income prices at the top of the schedule	37.625% against 37.625%	met
§7.2.2	$\sum_i \mathrm{EL}_i \le B_0$	at most 36.8% of B_0	met
§5.3, §7.2.1	at the contractual rate the payments are worth par	identity proved symbolically	met

6 Model Limitations and Known Weaknesses

1. **Benchmark, not validation.** The code is shown to compute its documented formulae, and the fair value to be the expectation of the lender's cashflows under the model's default process; whether an affordability-based spread is the right default process is a separate question.

2. **Downstream.** Every fair value with a loss given default rises, and with it the portfolio totals and the discount to par that consume it; figures produced before the fix are understated by up to the amounts above.

3. **Discounting.** At the risk-free rate the loan calculator supplies, a coupon above it values the loan above par, so the discount to par measures the coupon's margin as well as credit (v1.3, §5.3).

7 Document History

Date	Version	Description	Author
27-Oct-2026	0.1	First issue: tests v1.1; finds the loss counted twice	David K Kelly
27-Oct-2026	0.2	After the fix: tests v1.3	David K Kelly